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Subject : DSA

Activity: Fill in the Blanks

1. multiple structures / structures
2. array[size]
3. array[index].member
4. 5 students
5. student records / employee records
6. scanf()
7. emp[2].name
8. an array
9. &array[0]
10. loop (for loop)

Activity: Code Snippets – Array of Structures

1. 100 students `struct Student { int roll; char name[50]; }; struct Student students[100];`
2. 5 employees `struct Employee { int id; float salary; }; struct Employee emp[5]; for (int i = 0; i < 5; i++) scanf("%d %f", &emp[i].id, &emp[i].salary); for (int i = 0; i < 5; i++) printf("ID: %d Salary: %.2f\n", emp[i].id, emp[i].salary);`
3. 3 subjects for 10 students `struct Student { int marks[3]; }; struct Student students[10];`
4. Products `struct Product { char name[50]; float price; }; struct Product products[100];`
5. Cricket team `struct Player { char name[50]; int runs; }; struct Player team[11];`

Activity: Practice the Code – Array of Structures

```
#include <stdio.h> struct Car { char model[50]; int year; }; int main() { struct Car cars[3] = { {"Toyota", 2022}, {"Honda", 2023}, {"BMW", 2024} }; for (int i = 0; i < 3; i++) { printf("Model: %s, Year: %d\n", cars[i].model, cars[i].year); } return 0; }
```

Correction for Q9: The clean syntax for a pointer to the first element of a structure array is `struct StructName *ptr = array;` or `struct StructName *ptr = &array[0];`